Student's Assessment Number ……

SECTIOIN A (15 Marks)

Answer **all** questions in this section

1. Choose the correct answer from (i) to (x) and write its letter corresponding.

(i) What is the primary purpose of a firewall in computer security?

A To store files securely
B To prevent unauthorised access to or from a private network
C To increase the speed of your computer
D To fix software bugs

(ii) Which of the following is a good password practice?

A Use your name as your password
B Use a combination of letters, numbers, and symbols
C Use the same password for all accounts
D Share your password with your friends

(iii) Which of the following is an example of phishing?

A A hacker attempting to access your files directly
B An email that looks like it is from a trusted source asking you to provide sensitive information
C Installing a software update
D Using a VPN to protect your browsing activity

(iv) What is two-factor authentication (2FA)?

A A security system that uses two different passwords
B A system that requires two methods of identity verification to log into an account
C A software used to detect malware
D A tool for encrypting data

(v) What should you do if you receive a suspicious link in an email or text message?

A Click on it to check if it is safe
B Forward it to a friend
C Ignore it and delete the message
D Download the file attached to the message

(vi) What is the purpose of software updates in cybersecurity?

A To fix security vulnerabilities and improve the system's functionality
B To make your computer slower
C To uninstall unnecessary programs
D To change the appearance of your computer interface

(vii) How does the Internet support various online services like email, social media, and video streaming, apart from the World Wide Web?

A By providing underlying infrastructure and protocols for data exchange and communication.
B By only allowing text communication.
C By limiting services to social media platforms only.
D By focusing solely on online banking services.

(viii) What is an IP address, and why is it important for Internet communication?

A An IP address is a password used to access websites.
B An IP address is a unique identifier for devices that allows communication between them over the Internet.
C An IP address is a type of encryption used for securing data.
D An IP address is only needed for mobile data connections.

(ix) How do email services work, and why are they important for communication?

A Email services allow users to send and receive physical mail.
B Email services enable the digital exchange of messages and files, providing a fast and convenient communication method.
C Email services provide only chat-based communication.
D Email services are used solely for sending large media files.

(x) What is cloud computing, and how does it benefit users?

A Cloud computing stores data only on local hard drives.
B Cloud computing allows users to store, manage, and process data online without the need for physical storage devices.
C Cloud computing is limited to businesses and not useful for individuals.
D Cloud computing eliminates the need for Internet connectivity.

2. Match the items in **Column A** with the responses in **Column B** by writing the letter of the correct response below the item number in the table provided.

Column A	**Column B**
(i) Firewall (ii) Phishing (iii) Antivirus software (iv) Two-Factor Authentication (2FA) (v) Malware (vi) Encryption	A A security process where information is converted B Software designed to detect and remove viruses C A type of cyberattack where attackers impersonate legitimate organisations to trick individuals into providing sensitive information like usernames or passwords. D Malicious software designed to disrupt, damage, or gain unauthorised access to a computer system. into a coded form to prevent unauthorised access. and other malicious software from your computer. E A protective system that monitors and controls incoming and outgoing network traffic based on predetermined security rules. F A security feature that requires users to verify their identity through two methods, such as a password and a code sent to their phone. G Document Merging

ANSWERS

Column A	**(i)**	**(ii)**	**(iii)**	**(iv)**	**(v)**	**(vi)**
Column B						

SECTION B (70 marks)

Answer **all** questions in this section

3. Explain why

(a) The first generation computers were large, over-heating, and power inefficient.

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(b) The development of microprocessors was a major development in computer technology.

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(c) A mobile phone is regarded as a computer. (Three reasons)

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(d) Most people prefer using laptop over desktop computer. (Three reasons)

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4. You have been assigned by the Computer Science (CS) teacher to prepare the students' magazine and submit it to his/her email addresses. When composing it, three fields named **To field**. **CC field**, and **BCC field** appeared.

   (a) Why the teacher opted such way of communication? Briefly explain by giving three reasons.

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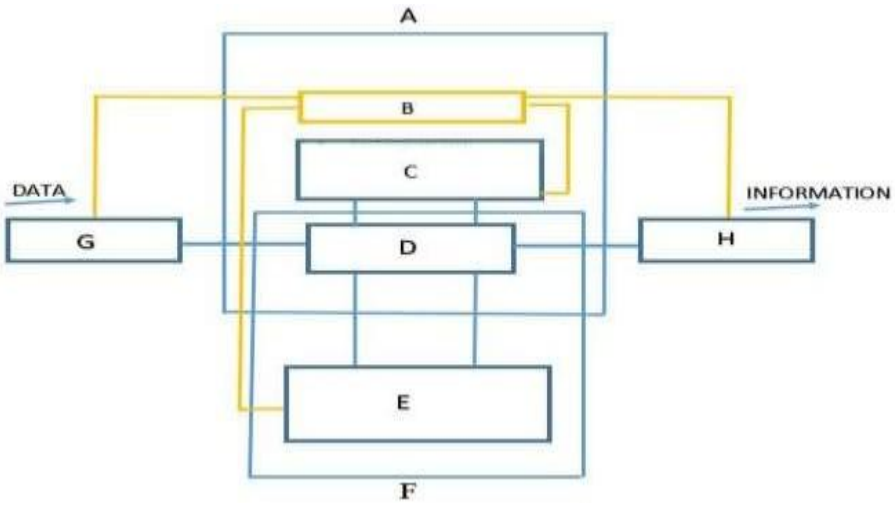
(b) Explain the function of each field appeared during composing an e-mail.

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(c) Why is it important to use CC and BCC fields appropriately when sending emails?

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5. Study carefully the figure below which shows a block diagram of typical computer systems, then answer the questions follows.



(a) Name the part which indicated by letters A to H

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(b) Explain the functions of parts labeled by letters B and C.

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(c) The motherboard is the main circuit board in the computer. Explain the steps followed to install the components labeled by letter "A" on motherboard of the desktop computer.

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6. (a) What are advantages and disadvantages of using solid state drives (SSDs) compared to hard disk drives (HDDs) for data storage?

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(b) Dickson is a Form Two student at Kimbiji Secondary School. One day, his Computer Science teacher, Sir Damas, instructed him to install a new software application on a computer. Outline five important factors that Dickson should understand or consider before installing the software.

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7. (a) Define the term "Computer handling and care". What do you think the importance of computer handling and care? Give three points

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(b) Outline four functions of Uninterruptible Power Supplies (UPS)

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8. Imagine you are computer technician at school. Explain shortly the steps you would take to resolve the following problems if;
   (a) A printer is not responding to print requests.

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   (b) An external hard drive is not showing up on the computer.

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   (c) A Computer monitor displays a distorted or no image at all.

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   (d) A student informs you that their computer run slower than usual, and they are unable to save their work.

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9. Your friend uses public Wi-Fi to log in to social media and Later, he discovers that his social media account has been hacked.
   (a) Explain the three possible ways through which his account might have been hacked when using public Wi-Fi.

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(b) Describe the four measures he should take to protect his social media account when using internet services in public places.

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SECTION C (15 Marks)

Answer question **ten (10)**

10. The world is proceeding to paperless and cashless transactions. For example, the general manager of BENICO company, Kole, always requests and makes all payments online. Kole checks the company's bank balance using a mobile phone, enters delivery address and payment details. The computer system then takes steps to ensure transaction safety by confirming that the person accessing the system is the authorized user. In four (4) points, explain how the computer system confirms that Kole is the authorized person to access and perform transactions.

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